



Solution of Midterm II

Course: Electromagnetic Theory

Instructor: Dr. Shafayat Abrar

Marks: 75 points [20% of the total]

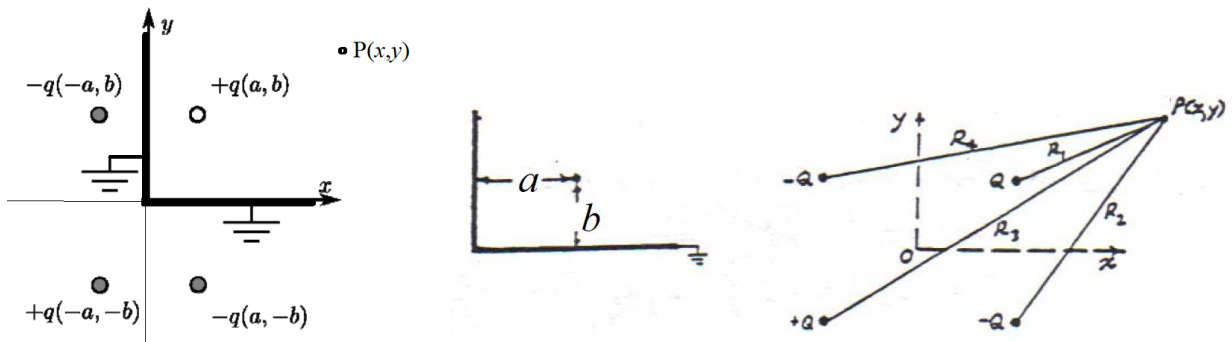
Date: April 06, 2018

Term: Spring 18

Duration: 1.5 hours

1. **[25 points]** There is a point charge $+q$ at coordinate (a, b) in front of an (infinitely long) **L-shaped** grounded conducting plane as shown in figure below. Exploiting method of images, we consider three image charges.

- Find electric potential V at field point $P(x, y)$ in the first quadrant.
- Find electric field by taking gradient of V .
- To calculate charge density σ_y on the vertical surface (y -axis), set $x = 0$ in the expression of electric field, and apply suitable boundary condition to find σ_y .
- To calculate charge density σ_x on the horizontal surface (x -axis), set $y = 0$ in the expression of electric field, and apply suitable boundary condition to find σ_x .



Solution: It is simple to obtain potential V at point P as a superposition of four point charges as follows:

$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} + \frac{1}{R_3} - \frac{1}{R_4} \right)$$

where

$$R_1 = \sqrt{(x-a)^2 + (y-b)^2}, \quad R_2 = \sqrt{(x-a)^2 + (y+b)^2}$$

$$R_3 = \sqrt{(x+a)^2 + (y+b)^2}, \quad R_4 = \sqrt{(x+a)^2 + (y-b)^2}$$

Now obtain $\mathbf{E}$ as follows:

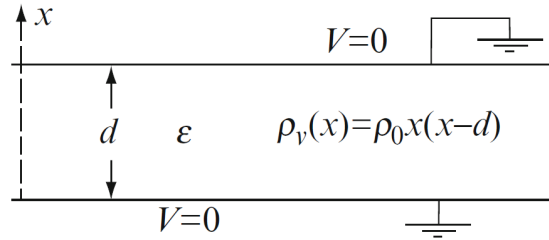
$$\mathbf{E} = -\hat{\mathbf{x}} \frac{\partial}{\partial x} V - \hat{\mathbf{y}} \frac{\partial}{\partial y} V$$

2. **[25 points]** Consider a line charge (infinite in length) with density λ located at height h above the grounded conducting plane.

- Using method of images and exploiting superposition, find electric field $\mathbf{E}$ at some field point $P(x, y, z)$ above the ground (due to both original and image line charges).
- Evaluating line integration of $\mathbf{E}$, obtain electric potential $V(x, y, z)$.
- Draw field lines above the ground (depicting clearly how they emerge out of line charge and terminate on conducting plane).
- Using boundary condition, obtain surface charge density.
- Show that the line integral of surface charge density is equal to image line charge.

Solution: It is solved in Sadiku; refer to page 243.

3. **[15+10 points]** The plates of a capacitor are separated by a distance d [m] and grounded as in figure below. The dielectric between the plates has permittivity ϵ [F/m], and a charge density $\rho_v(x) = \rho_0 x(x - d)$ [C/m³] is distributed throughout the volume of dielectric. Solving *Poisson's equation*, find potential and electric field everywhere between the plates of capacitor.



Solution: Assuming homogeneous medium, solve the Poisson's equation

$$\nabla^2 V = -\frac{\rho_v}{\epsilon} \Rightarrow \frac{d^2}{dx^2} V(x) = -\frac{\rho_0}{\epsilon} x(x - d)$$

Integrate this expression twice, you get an expression for $V(x)$ with two integration constants A and B . Use the boundary conditions $V(x = 0) = 0$ and $V(x = d) = 0$. Once you have the expression of V , find $\mathbf{E} = -\nabla V = -\left(\frac{d}{dx} V\right) \hat{\mathbf{x}}$.

4. **[15 points]** For a z -directed dipole, show that the electric field is given by

$$\mathbf{E}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos(\theta) \hat{\mathbf{r}} + \sin(\theta) \hat{\boldsymbol{\theta}})$$

Solution: The derivation of electric potential $V(r, \theta)$ was explained in the class. Obtain $V(r, \theta)$, and evaluate $\mathbf{E} = -\nabla V$ in spherical coordinate system.

